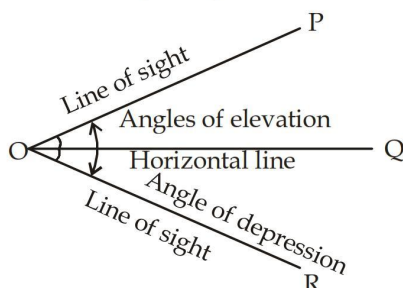


Height and Distance

Height and distance is one of the important application of trigonometry. It is a technique used to find the distance and height of the object. We can find them with the help of trigonometric ratios.

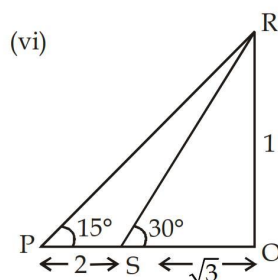
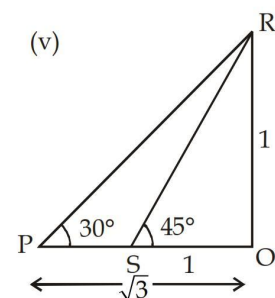
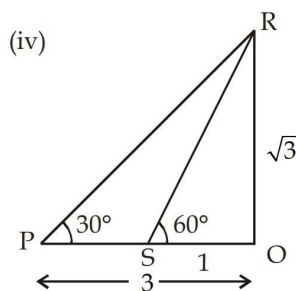
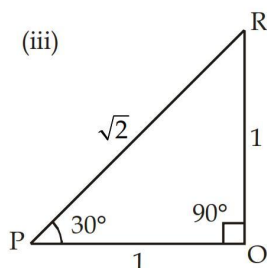
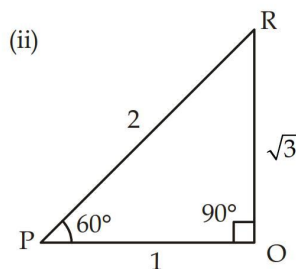
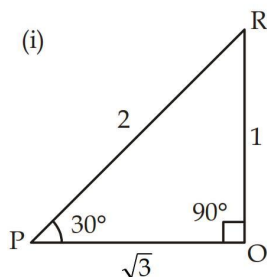


Line of Sight : The line of sight is the line drawn from the eye of an observer to the object.

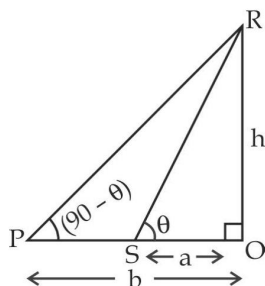
Angle of Elevation : When the object is above the horizontal level of our eye, we have to turn our head upwards to see an object. Here $\angle POQ$ is the angle of elevation.

Angle of Depression : When the object is below the horizontal level of our eye, we have to turn our head downwards to see an object. Here $\angle ROQ$ is the angle of depression.

Some Important Points : In this chapter we solve all the questions with the help of ratio.



Concept 1 : If the angle of elevation of the top of a tower at two points which are at a distance of 'a' and 'b' metres from the foot of tower and on the same side of the tower are complementary. Then height of the tower is $\sqrt{ab}$.



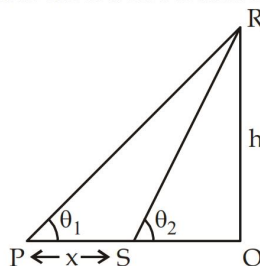
Example : If the angle of elevation of the top of a tower at two points which are at a distance of '9' and '4' metres from the foot of the tower and on the same side of the tower are complementary. Find the height of the tower?

Sol. Height of tower = $\sqrt{ab} = \sqrt{9 \times 4}$
 $= \sqrt{36} = 6 \text{ m}$

Concept 2 : If a man wishes to find the height of a tower which stands on a horizontal plane. The angle of elevation of top of the tower is θ_1 . On walking x units towards the tower. He find the angle of elevation becomes θ_2 . Then the height of the tower is

$$h = \left[\frac{x \tan \theta_1 \tan \theta_2}{\tan \theta_2 - \tan \theta_1} \right]$$

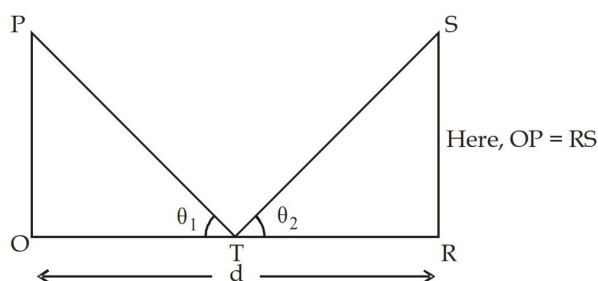
$$h = \frac{x}{\cot \theta_1 - \cot \theta_2}$$



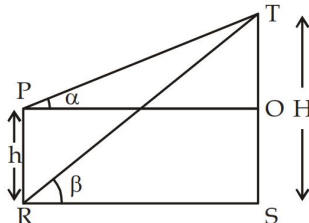
Concept 3 : If two poles of equal heights stand on either sides of a road which is 'd' units wide. At a point on the road between the poles, the elevation of the top of the poles are θ_1 and θ_2 , then height of the poles is

$$h = \left[\frac{d \tan \theta_1 \tan \theta_2}{\tan \theta_1 + \tan \theta_2} \right]$$

$$h = \frac{d}{\cot \theta_1 + \cot \theta_2}$$



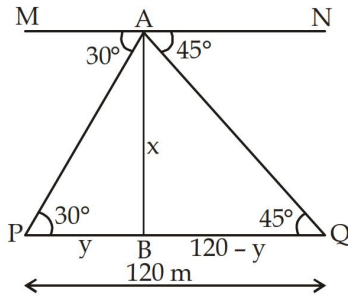
Concept 4 : From the top and bottom of the building of height 'h' units, the angle of elevation of the top of a tower are ' α ' and ' β ' respectively, then the height of the tower is $\left[\frac{h \tan \beta}{\tan \beta - \tan \alpha} \right]$



Types of Questions

1. The angles of depression of two ships from the top of a lighthouse are 45° and 30° . If the ships are 120 m apart, find the height of the lighthouse? (Ships are on both side of light house)

Sol. Let AB, the height of the lighthouse = x m



Since $MN \parallel PQ$

$\therefore$ angle $MAP =$ angle $APB = 30^\circ$ and angle

$NAQ =$ angle $AQB = 45^\circ$

Let the length between P and B be y m.

So, the length between B and Q is $(120 - y)$ m.

In $\triangle ABP$

$$\tan 30^\circ = \frac{AB}{BP} \Rightarrow \frac{1}{\sqrt{3}} = \frac{x}{y}$$

$$\Rightarrow y = x\sqrt{3} \quad \dots(i)$$

Again, in $\triangle ABQ$

$$\tan 45^\circ = \frac{AB}{BQ} \Rightarrow 1 = \frac{x}{120 - y}$$

$$\Rightarrow x = 120 - y \quad \dots(ii)$$

Combining equations (i) and (ii), we get

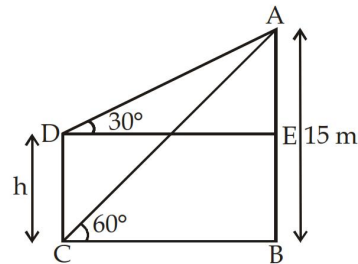
$$x = 120 - x\sqrt{3}$$

$$\text{or, } x(1 + \sqrt{3}) = 120$$

$$\therefore x = \frac{120}{1 + \sqrt{3}} \approx 44 \text{ m}$$

2. The top of a 15 metre-high tower makes an angle of depression of 60° with the bottom of an electric pole and an angle of depression of 30° with the top of the pole. What is the height of the electric pole?

Sol. $15 = \frac{h(\tan 60^\circ)}{\tan 60^\circ - \tan 30^\circ}$

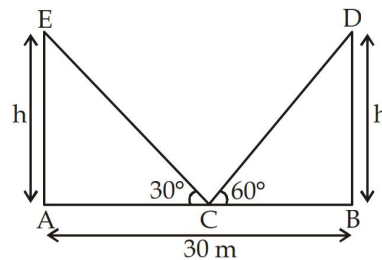


$$\therefore h = \frac{15 \times (\tan 60^\circ - \tan 30^\circ)}{\tan 60^\circ} = \frac{15 \times \left(\sqrt{3} - \frac{1}{\sqrt{3}} \right)}{\sqrt{3}}$$

$$= \frac{30}{\sqrt{3} \times \sqrt{3}} = 10 \text{ m}$$

3. Two poles of equal heights are standing opposite to each other on either side of a road, which is 30 m wide. From a point between them on the road, the angles of elevation of the tops are 30° and 60° . The height of each pole is _____?

Sol. Required height (h) = $\frac{30 \times \tan 30^\circ \times \tan 60^\circ}{\tan 60^\circ + \tan 30^\circ}$



$$= \frac{30 \times \frac{1}{\sqrt{3}} \times \sqrt{3}}{\sqrt{3} + \frac{1}{\sqrt{3}}} = \frac{15\sqrt{3}}{2} = 7.5 \times 1.732$$

$$= 12.975 \text{ m} \approx 13 \text{ m}$$

4. The angle of elevation of a lamppost changes from 30° to 60° when a man walks towards it. If the height of the lamppost is $10\sqrt{3}$ metres, find the distance travelled by men?

Sol. Height of tower = $\frac{\text{Distance travelled}}{\cot \theta_1 - \cot \theta_2}$

$$h = \frac{x}{\cot \theta_1 - \cot \theta_2}, \quad 10\sqrt{3} = \frac{x}{\cot 30^\circ - \cot 60^\circ}$$

$$x = 10\sqrt{3} \left(\frac{2}{\sqrt{3}} \right) = 20 \text{ m}$$

Foundation

1. The height of a pole and the shadow cast by it have equal lengths. The angle of elevation of the sun is:
 - (a) 30°
 - (b) 60°
 - (c) 45°
 - (d) 75°
2. A ladder of length 20 m is resting against a wall 10 m high such that the top of the ladder touches the top edge of the wall. Find the distance of the foot of the ladder from the foot of the wall?
 - (a) $\frac{10}{\sqrt{3}}$ m
 - (b) 10 m
 - (c) $10\sqrt{3}$ m
 - (d) 30 m
3. A ladder is resting against a wall such that the top of the ladder and that of the wall touch each other. The height of the wall is $\sqrt{3}$ times the distance of the foot of the ladder from the foot of the wall. The angle which the ladder makes with the ground is:
 - (a) 30°
 - (b) 60°
 - (c) 45°
 - (d) 75°
4. The angle of elevation of a moon when the length of the shadow of a pole is equal to its height, is:
 - (a) 30°
 - (b) 45°
 - (c) 60°
 - (d) 90°
5. If the length of shadow of a pole on a level ground is $\sqrt{3}$ times the length of that pole, the angle of elevation of the sun is:
 - (a) 30°
 - (b) 45°
 - (c) 60°
 - (d) None of these
6. The angle of elevation of a tower from a distance 100 m from its foot is 30° . Height of the tower is:
 - (a) $100\sqrt{3}$ m
 - (b) $\frac{100}{\sqrt{3}}$ m
 - (c) $50\sqrt{3}$ m
 - (d) $\frac{200}{\sqrt{3}}$ m
7. When the length of the shadow of a pillar on the ground is same as the height of the pillar then the angle of elevation of the sun is:
 - (a) $\frac{\pi}{2}$
 - (b) $\frac{\pi}{3}$
 - (c) $\frac{\pi}{6}$
 - (d) $\frac{\pi}{4}$
8. The length of a shadow of a vertical tower is $\frac{1}{\sqrt{3}}$ times its height. The angle of elevation of the sun is:
 - (a) 30°
 - (b) 45°
 - (c) 60°
 - (d) 90°
9. The length of the shadow of a vertical pole 9 m high, when the sun's altitude is 30° , is (in cm):
 - (a) $3\sqrt{3}$
 - (b) 9
 - (c) $9\sqrt{3}$
 - (d) $18\sqrt{3}$
10. P and Q are two points observed from the top of a building $10\sqrt{3}$ m high. If the angles of depression of the points are complementary and $PQ = 20$ m, then the distance of P from the building is: (P being the farther point)
 - (a) 25 m
 - (b) 45 m
 - (c) 30 m
 - (d) 40 m
11. If angle of elevation of top of a tower AB from a point P on the ground 15 meter away from the foot is 60° , find the height of the tower?
 - (a) $\frac{15}{\sqrt{3}}$ m
 - (b) 15 m
 - (c) $15\sqrt{3}$ m
 - (d) 30 m
12. Length of the tight thread of a kite from a point on the ground is 85 m. If thread subtends an angle θ with the ground such that $\tan \theta = \frac{8}{15}$ then find the height at which kite is flying?
 - (a) 20 m
 - (b) 40 m
 - (c) 35 m
 - (d) 50 m
13. To cross a river a person covers a straight forward distance of 325 m along a bridge over the river. If bridge subtends 30° angle with edge of the river, find the width of the river?
 - (a) 152.5
 - (b) 155
 - (c) 165
 - (d) 162.5
14. In a storm, a tree got broken by the wind whose top meets the ground at an angle of 30° , at a distance of 30 meters from the root. What was the height of the tree before breaking?
 - (a) 30 m
 - (b) $30\sqrt{3}$ m
 - (c) 60 m
 - (d) $60\sqrt{3}$ m
15. The height of a tower is 50 meter. When the Sun's altitude increases from 30° to 45° , the length of the shadow of the tower is decreased by x meter. Find the approximate value of x in meter?
 - (a) 36.6 m
 - (b) 35.6 m
 - (c) 86.6 m
 - (d) 36 m
16. The shadow of a tree is 16 meter when elevation of Sun is 60° . What is the height of the tree?
 - (a) 8 m
 - (b) 16 m
 - (c) $16\sqrt{3}$ m
 - (d) $\frac{16}{\sqrt{3}}$ m

17. A national highway is in north-south direction. From a point on the highway a school is situated in the north-east direction and is at a distance of $600\sqrt{2}$ m from the point. What is the perpendicular distance between school and the highway?
- (a) 600 m (b) $\frac{600}{\sqrt{3}}$ m
(c) $600\sqrt{3}$ m (d) 300 m
18. When Sun's altitude is 30° , the shadow of a tower is 15 meter. What is the length of shadow when Sun's altitude is 60° ?
- (a) 3 m (b) 4 m
(c) 5 m (d) 6 m
19. The shadow of a 6 m high tower is 15 m and at the same point of time length of shadow of a tree is 25 m. What is the height of the tree?
- (a) 21 m (b) 10 m
(c) 35 m (d) None of these
20. A tower stands vertically on the ground. From a point on the ground which is 30 m away from the foot of a tower, the angle of elevation of the top of the tower is found to be 45° . Find the height of the tower.
- (a) 15 (b) 40
(c) 30 (d) 20
21. A tower is $50\sqrt{3}$ meters high. Find the angle of elevation of its top from a point 50 meters away from its foot?
- (a) $\theta = 60^\circ$ (b) $\theta = 45^\circ$
(c) $\theta = 30^\circ$ (d) $\theta = 22\frac{1}{2}^\circ$
22. The angle of elevation of the top of a tower at a distance of 30 m from its foot is 60° . The height of the tower is:
- (a) 20 m (b) $30\sqrt{3}$ m
(c) $15\sqrt{2}$ m (d) $\frac{15}{\sqrt{2}}$ m
23. If the ratio of the length of a pen to its shadow is $1 : \sqrt{3}$, the angle of elevation of the source of light is:
- (a) 40° (b) 30°
(c) 60° (d) 90°
24. The angle of elevation of the top of a tower at a distance of 500 m from its foot is 30° . The height of the tower is:
- (a) $\frac{500(\sqrt{3}-1)}{3}$ m (b) 500 m
(c) $\frac{500\sqrt{3}}{3}$ m (d) $\frac{500(\sqrt{3}+1)}{3}$
25. If the angle of elevation of the top of a building from a point 50 m away from its base is 60° , the height of the building is:
- (a) $100\sqrt{3}$ m (b) $50(\sqrt{3}-1)$ m
(c) $\frac{50}{\sqrt{3}}$ m (d) $50\sqrt{3}$ m
26. If the angle of elevation of sun is θ and the length of the shadow of a pole of length P is S, then:
- (a) $P = S \cos \theta$ (b) $P = S \sin \theta$
(c) $P = \frac{S}{\cot \theta}$ (d) $P = S \cot \theta$
27. Determine the length of a ladder, if it is leaning against a vertical wall making an angle of inclination of 30° with the ground and its foot is 15 m from the wall?
- (a) 15.77 m (b) 10 m
(c) $10\sqrt{3}$ m (d) 12.71 m
28. The foot of a ladder leaning against a wall of length 5 metre rest on a level ground $5\sqrt{3}$ metre from the base of the wall. The angle of inclination of the ladder with the ground is:
- (a) 60° (b) 50°
(c) 40° (d) 30°
29. A balloon is connected to a meteorological station by a cable of length 200 m, inclined at 60° to the horizontal. Find the height of the balloon from the ground. Assume that there is no slack in the cable?
- (a) 173.2 m (b) 17.35 m
(c) 123.2 m (d) None of these
30. The length of a string between a kite and a point on the ground is 50 m. The string makes an angle of 60° with the level ground. If there is no slack in the string, the height of the kite is:
- (a) $50\sqrt{3}$ m (b) $25\sqrt{3}$ m
(c) 25 m (d) $\frac{25}{\sqrt{3}}$ m

Moderate

1. The angle of elevation of the top of a tower from two points situated at distances 36 m and 64 m from its base and in the same straight line with it are complementary. What is the height of the tower?
 (a) 50 m (b) 48 m
 (c) 25 m (d) 24 m
2. The angle of elevation of the top of a tower standing on a horizontal plane from two points on a line passing through the foot of the tower at a distance 9 ft and 16 ft respectively are complementary angles. Then the height of the tower is:
 (a) 9 ft (b) 12 ft
 (c) 16 ft (d) 144 ft
3. At a point on a horizontal line through the base of a monument the angle of elevation of the top of the monument is found to be such that its tangent is $\frac{1}{5}$. On walking 138 metres towards the monument the secant of the angle of elevation is found to be $\frac{\sqrt{193}}{12}$. The height of the monument (in metre) is:
 (a) 35 (b) 49
 (c) 42 (d) 56
4. The angles of elevation of the top of a tower from two points A and B lying on the horizontal through the foot of the tower are respectively 15° and 30° . If A and B are on the same side of the tower and $AB = 48\text{m}$, then the height of the tower is:
 (a) $24\sqrt{3}$ m (b) 24 m
 (c) $24\sqrt{2}$ m (d) 96 m
5. The angles of depression, from the top of a light-house, of two boats are 45° and 30° towards the west. If the two boats are 6m apart, then the height of the light-house is:
 (a) $3(\sqrt{3} + 1)\text{m}$ (b) $(\sqrt{3} + 1)\text{m}$
 (c) $3(\sqrt{3} - 1)\text{m}$ (d) $(\sqrt{3} - 1)\text{m}$
6. The angle of elevation of the top of an incomplete vertical pillar at a horizontal distance of 100 m from its base is 45° . If the angle of elevation of the top of complete pillar at the same point is to be 60° , then the height of the incomplete pillar is to be increased by:
 (a) $50\sqrt{2}$ (b) 100 m
 (c) $100(\sqrt{3} - 1)\text{m}$ (d) $100(\sqrt{3} + 1)\text{m}$
7. A flag staff 5 mt. high stands on a building 25 mt. high. An observer is at the height of 30 m. The flag staff and the building subtend equal angles. The distance of the observer from the top of the flag staff is:
 (a) $\frac{5\sqrt{3}}{\sqrt{8}}$ (b) $\frac{5\sqrt{3}}{\sqrt{2}}$
 (c) $\frac{4\sqrt{3}}{\sqrt{2}}$ (d) $\frac{7\sqrt{3}}{\sqrt{2}}$
8. If from the top of a tower 50 m high, the angles of depression of two objects are respectively 60° and 45° , then the approximate distance between the objects, is: (objects are on same side of tower)
 (a) 11 m (b) 21 m
 (c) 31 m (d) 41 m
9. From the top of a pillar of height 20 m, the angles of elevation and depression of the top and bottom of another pillar are 30° and 45° respectively. The height of the second pillar (in meters) is:
 (a) $\frac{20(\sqrt{3} + 1)}{\sqrt{3}}$ (b) 10
 (c) $10\sqrt{3}$ (d) $\frac{10}{\sqrt{3}}(\sqrt{3} + 1)$
10. A straight tree breaks due to storm and the broken part bends so that the top of the tree touches the ground making an angle of 30° with the ground. The distance from the foot of the tree to the point where the top touches the ground is 10 metres. The height of the tree is:
 (a) $10(\sqrt{3} + 1)\text{m}$ (b) $10\sqrt{3}\text{m}$
 (c) $10(\sqrt{3} - 1)\text{m}$ (d) $\frac{10}{\sqrt{3}}\text{m}$
11. If the angles of elevation of a tower from two points distant a and b ($a > b$) from its foot and in the same straight line from it are 30° and 60° , then the height of the tower is:
 (a) $\sqrt{a + b}$ (b) $\sqrt{ab}$
 (c) $\sqrt{a - b}$ (d) $\sqrt{\frac{a}{b}}$

12. A person of height 2 m wants to get a fruit which is on a pole of height $\left(\frac{10}{3}\right)$ m. If he stands at a distance of $\left(\frac{4}{\sqrt{3}}\right)$ m from the foot of the pole, then the angle at which he should throw the stone, so that it hits the fruit, is:
 (a) 15° (b) 30°
 (c) 45° (d) 60°
13. A landmark on a river bank is observed from two points A and B on the opposite bank of the river. The lines of sight make equal angles with the bank of the river. If $AB = 1$ km, then the width of the river is:
 (a) 2 km (b) $\frac{1}{2}$ km
 (c) $\frac{3\sqrt{2}}{2}$ km (d) $\frac{\sqrt{3}}{2}$ kms
14. A, B, C are three collinear points on the ground such that B lies between A and C and $AB = 10$ m. If the angles of elevation of the top of a vertical tower standing at C are respectively 30° and 60° as seen from A and B, then the height of the tower is:
 (a) $5\sqrt{3}$ m (b) 5 m
 (c) $\frac{10\sqrt{3}}{3}$ m (d) $\frac{20\sqrt{3}}{3}$ m
15. The angles of elevation of the top of a tower from two points A and B lying on the horizontal plane through the foot of the tower are respectively 45° and 30° . If A and B are on the same side of the tower and $AB = 48$ metre, then the height of the tower is:
 (a) $24(\sqrt{3} + 1)$ metre (b) 24 metre
 (c) $14\sqrt{2}$ metre (d) 96 metre
16. There are two vertical posts, one on each side of a road, just opposite to each other. One post is 108 metre high. From the top of this post, the angles of depression of the top and foot of the other post are 30° and 60° respectively. The height of the other post, in metre, is:
 (a) 36 (b) 72
 (c) 108 (d) 110
17. The angles of elevation of the top of a building and the top of the chimney on the roof of the building from a point on the ground are x and 45° respectively. The height of building is h metre. Then the height of the chimney, in metre, is:
 (a) $h \cot x + h$ (b) $h \cot x - h$
 (c) $h \tan x - h$ (d) $h \tan x + h$
18. The angle of elevation of the top of a tower from a point A on the ground is 30° . On moving a distance of 20 metres towards the foot of the tower to a point B, the angle of elevation increases to 60° . The height of the tower is:
 (a) $\sqrt{3}$ m (b) $5\sqrt{3}$ m
 (c) $10\sqrt{3}$ m (d) $20\sqrt{3}$ m
19. A telegraph post is bent at a point above the ground due to storm. Its top just meets the ground at a distance of $8\sqrt{3}$ metres from its foot and makes an angle of 30° , then the height of the post is:
 (a) 16 metres (b) 23 metres
 (c) 24 metres (d) 10 metres
20. A ladder is resting against a wall at a height of 10 m. If the ladder is inclined with the ground at an angle of 30° , then the distance of the foot of the ladder from the wall is:
 (a) $\frac{10}{\sqrt{3}}$ m (b) $\frac{20}{\sqrt{3}}$ m
 (c) $10\sqrt{3}$ m (d) $20\sqrt{3}$ m

Difficult

1. At the base of a mountain the elevation of its summit is 45° . After ascending 600 meter towards the mountain upon an incline of 30° , the elevation changes to 60° . Find the height of the mountain?
 (a) 300 m (b) $300\sqrt{3}$ m
 (c) $300(1 + \sqrt{3})$ m (d) $300(\sqrt{3} - 1)$ m
2. The angle of elevation of an aeroplane from a point on the ground is 60° . After flying for 15 seconds, the elevation changes to 30° . If the aeroplane is flying at a speed of 720 kmph, then find the constant height at which aeroplane is flying?
 (a) $\frac{\tan \beta - \tan \alpha}{a \tan \alpha \tan \beta}$ (b) $\frac{a(\tan \beta - \tan \alpha)}{\tan \alpha \tan \beta}$
 (c) $\frac{a \tan \alpha \tan \beta}{\tan \alpha - \tan \beta}$ (d) $\frac{a \tan \alpha \tan \beta}{\tan \beta - \tan \alpha}$
3. From a point on the ground the angle of elevation of top of a tower is α . On moving 'a' meters towards the tower, the elevation changes to β . The height of the tower is:
 (a) 2958 m (b) 3850 m
 (c) 2598 m (d) 3598 m

4. An aeroplane is flying above a horizontal plane. The angles of depression of two consecutive mile stones from plane in opposite directions are α and β . The aeroplane is flying at a height of:
- (a) $\frac{\tan \alpha \tan \beta}{\tan \alpha - \tan \beta}$ (b) $\frac{\tan \alpha - \tan \beta}{\tan \alpha \tan \beta}$
- (c) $\frac{\tan \alpha \tan \beta}{\tan \alpha + \tan \beta}$ (d) $\frac{\tan \alpha + \tan \beta}{\tan \alpha \tan \beta}$
5. The angle of elevation of a stationary cloud from a point h meter above a lake is α and angle of depression of its reflection in the lake is β . The height of the cloud above the lake is:
- (a) $\frac{h(\tan \beta + \tan \alpha)}{\tan \alpha - \tan \beta}$ (b) $\frac{h(\tan \beta - \tan \alpha)}{\tan \beta + \tan \alpha}$
- (c) $\frac{h(\tan \alpha - \tan \beta)}{\tan \alpha + \tan \beta}$ (d) $\frac{h(\tan \beta + \tan \alpha)}{\tan \beta - \tan \alpha}$
6. A spherical balloon of radius a units subtends an angle θ , at an observer's eye. The angle of elevation of centre of the balloon is ϕ . The height of the centre of the ball is:
- (a) $a \sin \theta \operatorname{cosec} \frac{\phi}{2}$ (b) $a \sin \frac{\theta}{2} \operatorname{cosec} \phi$
- (c) $a \sin \phi \operatorname{cosec} \frac{\theta}{2}$ (d) $a \sin \frac{\phi}{2} \operatorname{cosec} \theta$
7. From vertically situated aeroplane to the straight horizontal road, the angle of depression of two consecutive km stones are α and β . If an aeroplane is in vertical plane in between two stones, then the height of the aeroplane from the road (in km) will be:
- (a) $\frac{\tan \alpha \tan \beta}{\tan \alpha + \tan \beta}$ (b) $\frac{\tan \alpha \tan \beta}{\tan \alpha - \tan \beta}$
- (c) $\frac{\tan \alpha + \tan \beta}{\tan \alpha - \tan \beta}$ (d) $\frac{\tan \alpha - \tan \beta}{\tan \alpha + \tan \beta}$
8. Each side of an equilateral triangle subtends an angle of 60° at the top of a tower h m high located at the centre of the triangle. If a is the length of each side of the triangle, then:
- (a) $3a^2 = 2h^2$ (b) $2a^2 = 3h^2$
- (c) $a^2 = 3h^2$ (d) $3a^2 = h^2$
9. A round balloon of radius 10m subtends an angle 60° at the eye of an observer while the angle of elevation of its centre is 45° . The height of the centre of the balloon is:
- (a) $20\sqrt{2}$ (b) $20\sqrt{3}$
- (c) $10\sqrt{2}$ (d) $10\sqrt{3}$
10. A tower on horizontal ground leans towards the north. At two points due south at distance a and b respectively from the foot, the angular elevations of the top of the tower are α and β . Find the inclination θ of the tower to the horizontal?
- (a) $\frac{b \cot \alpha + a \cot \beta}{a - b}$ (b) $\frac{b \sin \alpha + b \cos \beta}{b - a}$
- (c) $\frac{b \cot \alpha - a \cot \beta}{b - a}$ (d) None of these
11. At the foot of the mountain the elevation of its summit is 45° ; after ascending 4 km towards the mountain up a slope of 30° inclination, the elevation is found to be 60° . Find the height of the mountain?
- (a) $2(\sqrt{3} + 1)$ km (b) $4(\sqrt{3} + 1)$ km
- (c) $2(\sqrt{3} - 1)$ km (d) $4(\sqrt{3} - 1)$ km
12. A boy standing in the middle of a field, observes a flying bird in the north at an angle of elevation of 30° and after 2 minutes, he observes the same bird in the south at an angle of elevation of 60° . If the bird flies all along in a straight line at a height of $50\sqrt{3}$ m, then its speed in km/h is:
- (a) 4.5 (b) 3
- (c) 9 (d) 6
13. A pole broken by the storm of wind and its top struck the ground at an angle of 30° and at a distance of 20 m from the foot of the pole, the height of the pole before it was broken:
- (a) $20\sqrt{3}$ m (b) $\frac{40\sqrt{3}}{3}$ m
- (c) $60\sqrt{3}$ m (d) $\frac{100\sqrt{3}}{3}$ m
14. The angle of depression of vertices of a regular hexagon lying in a plane from the top of a 75 m high tower standing at the centre of the hexagon is 60° . What is the length of each side of the hexagon?
- (a) $50\sqrt{3}$ m (b) 75 m
- (c) $25\sqrt{3}$ m (d) 25 m
15. Two vertical pillars stand on either side of a road. One of them is 108 meter height. From the top of this pillar angle of depression of top and bottom of the other pillar are respectively 30° and 60° . What is the height of the second pillar.
- (a) 36 m (b) 72 m
- (c) 108 m (d) 110 m

16. A spherical balloon of radius 10 feet is in the open air. If angle of elevation of centre of the balloon from a point on the ground is 45° and spherical balloon subtends an angle of 45° on that point then how high is the centre of the balloon from the ground?
 (a) 10 feet (b) 15 feet
 (c) $20\sqrt{2}$ feet (d) $10\sqrt{2}$ feet
17. The angle of elevation of an aeroplane from a point on the ground is 45° . After flying for 15 second, the elevation changes to 30° . If the aeroplane is flying at a height of 2500 meters, then approximate speed of the aeroplane in kmph is:
 (a) 440 kmph (b) 400 kmph
 (c) 360 kmph (d) 480 kmph
18. A boy standing in the middle of a field, observes a flying bird in the north at an angle of elevation of 30° and after 2 minutes, he observes the same bird in the south at an angle of elevation of 60° . If the bird flies all along in a straight line at a height of $50\sqrt{3}$ m, then its speed in km/h is:
 (a) 4.5 (b) 3
 (c) 9 (d) 6
19. The angle of elevation of an aeroplane from a point on the ground is 60° . After 15 seconds flight, the elevation changes to 30° . If the aeroplane is flying at a height of $1500\sqrt{3}$ m, then the speed of the plane is:
 (a) 300 m/sec (b) 200 m/sec
 (c) 100 m/sec (d) 150 m/sec
20. There are two temples, one on each bank of a river, just opposite to each other. One temple is 54 m high. From the top of this temple, the angles of depression of top and bottom of another temple are 30° and 60° respectively. The height of the other temple is:
 (a) 18 m (b) 36 m
 (c) $36\sqrt{3}$ m (d) $18\sqrt{3}$ m

Previous Year (Memory Based)

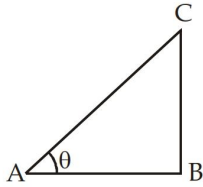
1. If the angle of elevation of the Sun changes from 30° to 45° , the length of the shadow of a pillar decreases by 20 metres. The height of the pillar is:
 (a) $20(\sqrt{3}-1)$ m (b) $20(\sqrt{3}+1)$ m
 (c) $10(\sqrt{3}-1)$ m (d) $10(\sqrt{3}+1)$ m
2. One flies a kite with a thread 150 metre long. If the thread of the kite makes an angle of 60° with the horizontal line, then the height of the kite from the ground (assuming the thread to be in a straight line) is:
 (a) 50 metre (b) $75\sqrt{3}$ metre
 (c) $25\sqrt{3}$ metre (d) 80 metre
3. The distance between two pillars of length 16 metres and 9 metres is x metres. If two angles of elevation of their respective top from the bottom of the other are complementary to each other, then the value of x (in metres) is:
 (a) 15 (b) 16
 (c) 12 (d) 9
4. The angles of elevation of the top of a building from the top and bottom of a tree are x and y respectively. If the height of the tree is h metre then, in metre, the height of the building is:
 (a) $\frac{h \cot x}{\cot x + \cot y}$ (b) $\frac{h \cot y}{\cot x + \cot y}$
 (c) $\frac{h \cot x}{\cot x - \cot y}$ (d) $\frac{h \cot y}{\cot x - \cot y}$
5. Two poles of equal heights are standing opposite to each other on either side of a road which is 100m wide. From a point between them on road, angles of elevation of their tops are 30° and 60° . The height of each pole in metre, is:
 (a) $25\sqrt{3}$ (b) $20\sqrt{3}$
 (c) $28\sqrt{3}$ (d) $30\sqrt{3}$
6. An aeroplane when flying at a height of 5000 m from the ground passes vertically above another aeroplane at an instant, when the angles of elevation of the two aeroplanes from the same point on the ground are 60° and 45° respectively. The vertical distance between the aeroplanes at that instant is:
 (a) $5000(\sqrt{3}-1)$ m (b) $5000(3-\sqrt{3})$ m
 (c) $5000\left(1-\frac{1}{\sqrt{3}}\right)$ m (d) 4500 m
7. A man standing at a point P is watching the top of a tower, which makes an angle of elevation of 30° . The man walks some distance towards the tower and then his angle of elevation of the top of the tower is 60° . If the height of the tower is 30 m, then the distance he moves is:
 (a) 22 m (b) $22\sqrt{3}$ m
 (c) 20 m (d) $20\sqrt{3}$ m

8. The distance between two vertical poles is 60 m. The height of one of the poles is double the height of the other. The angles of elevation of the top of the poles from the middle point of the line segment joining their feet are complementary to each other. The heights of the poles are:
 (a) 10 m and 20 m (b) 20 m and 40 m
 (c) 20.9 m and 41.8 m (d) $15\sqrt{2}$ m and $30\sqrt{2}$ m
9. The shadow of the tower becomes 60 metres longer when the altitude of the sun changes from 45° to 30° . Then the height of the tower is:
 (a) $20(\sqrt{3} + 1)$ m (b) $24(\sqrt{3} + 1)$ m
 (c) $30(\sqrt{3} + 1)$ m (d) $30(\sqrt{3} - 1)$ m
10. A vertical post 15 ft high is broken at a certain height and its upper part, not completely separated, meets the ground at an angle of 30° . Find the height at which the post is broken?
 (a) 10 ft (b) 5 ft
 (c) $15\sqrt{3}(2 - \sqrt{3})$ ft (d) $5\sqrt{3}$ ft
11. A man 6 ft tall casts a shadow 4 ft. long at the same time when a flag pole casts a shadow 50 ft long. The height of the flag pole is:
 (a) 80 ft (b) 75 ft
 (c) 60 ft (d) 70 ft
12. The angle of elevation of an aeroplane from a point on the ground is 60° . After 15 seconds flight, the elevation changes to 30° . If the aeroplane is flying at a height of $1500\sqrt{3}$ m, find the speed of the plane?
 (a) 300 m/sec (b) 200 m/sec
 (c) 100 m/sec (d) 150 m/sec
13. The angles of elevation of the top of a tower from the points P and Q at distances of 'a' and 'b' respectively from the base of the tower and in the same straight line with it are complementary. The height of the tower is:
 (a) $\sqrt{ab}$ (b) $\frac{a}{b}$
 (c) ab (d) a^2b^2
14. A pole stands vertically, inside a scalene triangular park ABC. If the angle of elevation of the top of the pole from each corner of the park is same, then in $\triangle ABC$, the foot of the pole is at the?
 (a) centroid (b) circumcentre
 (c) incentre (d) orthocentre
15. If the angles of elevation of a balloon from two consecutive kilometre-stones along a road are 30° and 60° respectively, then the height of the balloon above the ground will be:
 (a) $\frac{\sqrt{3}}{2}$ km (b) $\frac{1}{2}$ km
 (c) $\frac{2}{\sqrt{3}}$ km (d) $3\sqrt{3}$ km
16. A vertical stick 12 cm long casts a shadow 8 cm long on the ground. At the same time, a tower casts a shadow 40 m long on the ground. The height of the tower is:
 (a) 72 m (b) 60 m
 (c) 65 m (d) 70 m
17. The length of the shadow of a vertical tower on level ground increases by 10 meters when the altitude of the sun changes from 45° to 30° . Then the height of the tower is:
 (a) $5\sqrt{3}$ metre (b) $10(\sqrt{3} + 1)$ metre
 (c) $5(\sqrt{3} + 1)$ metre (d) $10\sqrt{3}$ metre
18. The elevation of the top of a tower from a point on the ground is 45° . On travelling 60 m from the point towards the tower, the elevation of the top becomes 60° . The height of the tower, in metres, is:
 (a) 30 (b) $30(3 - \sqrt{3})$
 (c) $30(3 + \sqrt{3})$ (d) $30\sqrt{3}$
19. The tops of two poles of height 24 m and 36 m are connected by a wire. If the wire makes an angle of 60° with the horizontal, then the length of the wire is:
 (a) 6 m (b) $8\sqrt{3}$ m
 (c) 8 m (d) $6\sqrt{3}$ m
20. From a tower 125 metres high, the angles of depression of two objects, which are in horizontal line through the base of the tower, are 45° and 30° and they are on the same side of the tower. The distance (in metres) between the objects is:
 (a) $125\sqrt{3}$ (b) $125(\sqrt{3} - 1)$
 (c) $\frac{125}{(\sqrt{3} - 1)}$ (d) $125(\sqrt{3} + 1)$

Foundation

Solutions

1. (c);



Let BC be the height of the pole and let AB be the length of its shadow.

According to the question:

$$BC = AB$$

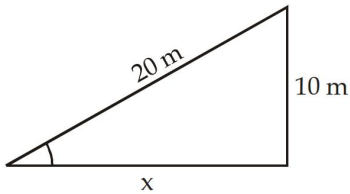
$$\Rightarrow \frac{BC}{AB}$$

$$\Rightarrow \tan \theta = \frac{BC}{AB} = 1$$

$$\Rightarrow \theta = 45^\circ$$

So, the angle of elevation of the sun is 45° .

2. (c);

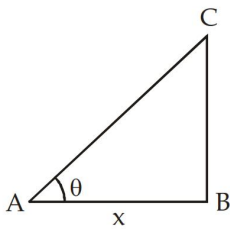


Let the required distance be x .

$$\sin \theta = \frac{10}{20} = \frac{1}{2}$$

$$\Rightarrow \theta = 30^\circ$$

$$\tan 30^\circ = \frac{10}{x} \Rightarrow \frac{1}{\sqrt{3}} = \frac{10}{x}, x = 10\sqrt{3}$$



3. (b);

AB = distance between the foot of the ladder and the foot of the wall.

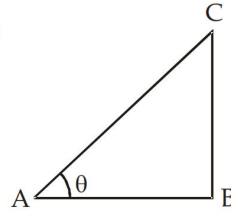
Let $AB = x$

$$\Rightarrow BC = \sqrt{3}x.$$

$$\tan \theta = \frac{\sqrt{3}x}{x} = \sqrt{3}$$

$$\Rightarrow \theta = 60^\circ$$

4. (b);



Let BC be the height of the pole and AB be the length of its shadow.

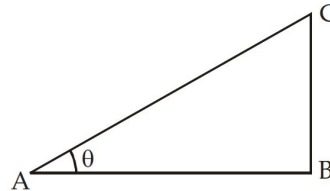
According to the question:

$$AB = BC$$

$$\Rightarrow \tan \theta = \frac{BC}{AB} = 1$$

$$\Rightarrow \theta = 45^\circ$$

5. (a);



Let BC be the length of the pole and let AB be the length of its shadow.

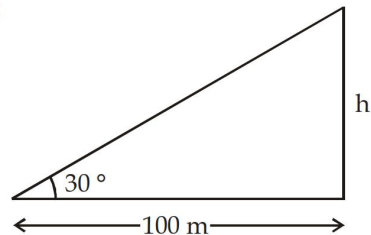
According to the question:

$$AB = \sqrt{3} BC$$

$$\tan \theta = \frac{BC}{AB} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \theta = 30^\circ$$

6. (b);

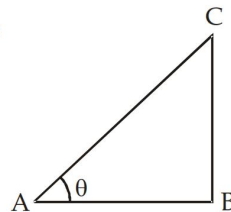


Let the h be the height of the tower.

$$h = 100 \tan 30^\circ$$

$$= \frac{100}{\sqrt{3}} \text{ m.}$$

7. (d);

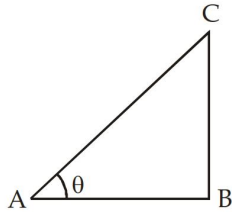


BC is the height of the pillar and AB is the length of the shadow.

$$\tan \theta = \frac{BC}{AB} = 1$$

$$\Rightarrow \theta = 45^\circ = \frac{\pi}{4} \text{ radians.}$$

8. (c);



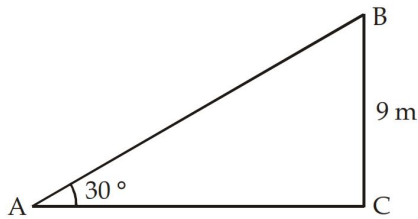
BC = height of the tower
AB = length of its shadow
According to the question:

$$AB = \frac{1}{\sqrt{3}} BC$$

$$\tan \theta = \frac{BC}{AB} = \sqrt{3}$$

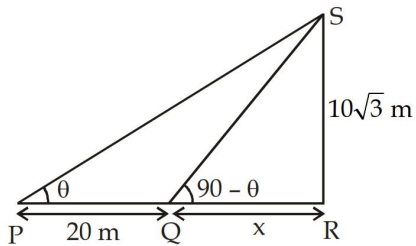
$$\Rightarrow \theta = 60^\circ$$

9. (c);



$$AC = 9 \cot 30^\circ = 9\sqrt{3} \text{ m.}$$

10. (c);



Let QR = x m.

$$\tan \theta = \frac{RS}{PR} = \frac{10\sqrt{3}}{20+x}$$

$$\tan (90 - \theta) = \cot \theta = \frac{10\sqrt{3}}{x}$$

$$\tan \theta \cdot \cot \theta = 1$$

$$\Rightarrow \frac{10\sqrt{3}}{20+x} \cdot \frac{10\sqrt{3}}{x} = 1$$

$$\Rightarrow x(20+x) = 300$$

$$x^2 + 20x - 300 = 0$$

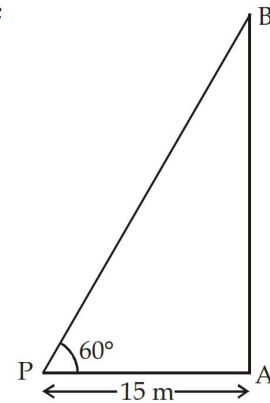
$$\Rightarrow x^2 + 30x - 10x - 300 = 0$$

$$\Rightarrow (x-10)(x+30) = 0$$

$$\Rightarrow x = 10 \text{ m.}$$

So, distance of P from the building = $20 + x = 30 \text{ m.}$

11. (c);

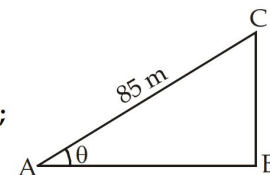


Let the height of the tower be h.

$$\frac{h}{15} = \tan 60^\circ$$

$$\Rightarrow h = 15\sqrt{3} \text{ m.}$$

12. (b);



$$\tan \theta = \frac{8}{15}$$

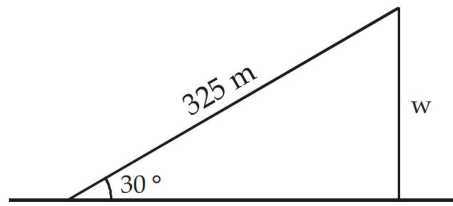
$$\sec \theta = \sqrt{1 + \tan^2 \theta} = \sqrt{1 + \frac{64}{225}} = \sqrt{\frac{225+64}{225}} = \frac{17}{15}$$

$$\Rightarrow \cos \theta = \frac{15}{17}$$

$$\Rightarrow \sin \theta = \sqrt{1 - \frac{225}{289}} = \sqrt{\frac{64}{289}} = \frac{8}{17} = \frac{BC}{85}$$

$$\Rightarrow BC = 40 \text{ m.}$$

13. (d);

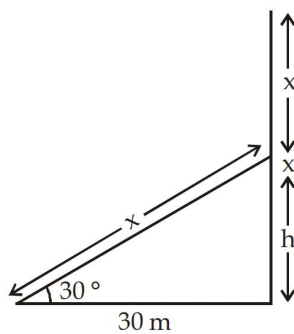


Let the width of the river be w .

$$\frac{w}{325} = \sin 30^\circ = \frac{1}{2}$$

$$\Rightarrow w = \frac{325}{2} = 162.5 \text{ m}$$

14. (b);



$$\frac{h}{30} = \tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$\Rightarrow h = \frac{30}{\sqrt{3}} \text{ m}$$

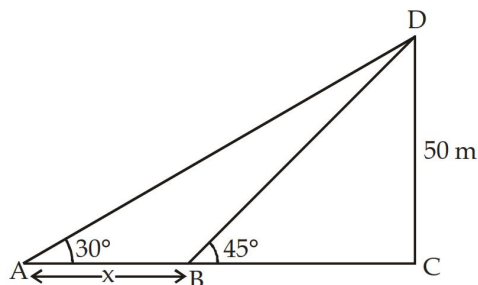
$$\frac{h}{x} = \sin 30^\circ = \frac{1}{2}$$

$$\Rightarrow x = 2h$$

So, height of the tree
 $= h + x = h + 2h = 3h$

$$= 3 \times \frac{30}{\sqrt{3}} \\ = 30\sqrt{3} \text{ m.}$$

15. (a);

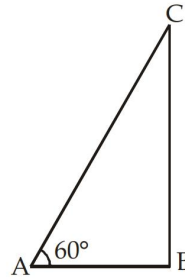


$$AC = 50 \cot 30^\circ = 50\sqrt{3} \text{ m}$$

$$BC = 50 \cot 45^\circ = 50 \text{ m}$$

$$x = AC - BC = 50(\sqrt{3} - 1) = 50 \times 0.732 = 36.6 \text{ m.}$$

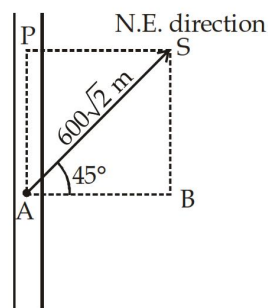
16. (c);



$$\frac{BC}{16} = \tan 60^\circ = \sqrt{3}$$

$$\Rightarrow BC = 16\sqrt{3} \text{ m.}$$

17. (a);



Perpendicular distance between the school and the highway

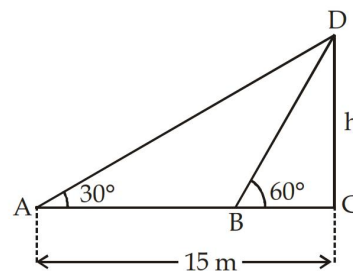
$PS = AB$

$$AB = 600\sqrt{2} \cos 45^\circ$$

$$= 600\sqrt{2} \times \frac{1}{\sqrt{2}} = 600 \text{ m.}$$

So, $PS = 600 \text{ m.}$

18. (c);



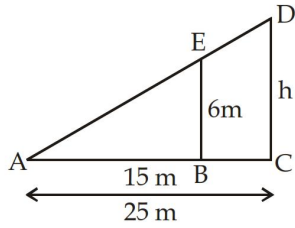
$$h = 15 \tan 30^\circ$$

$$= \frac{15}{\sqrt{3}} \text{ m.}$$

$$BC = h \cot 60^\circ$$

$$= \frac{15}{\sqrt{3}} \times \frac{1}{\sqrt{3}} = 5 \text{ m.}$$

19. (b);

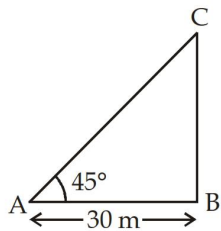


According to the question:

$$\frac{6}{15} = \frac{h}{25}$$

$$\Rightarrow h = 10 \text{ m.}$$

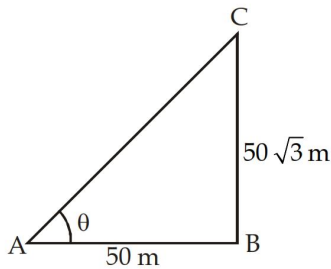
20. (c);



$$h = 30 \tan 45^\circ$$

$$= 30 \times 1 = 30 \text{ m.}$$

21. (a);

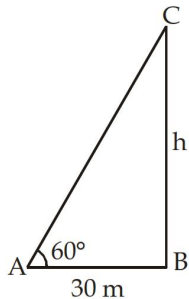


Let the angle of elevation be θ

$$\tan \theta = \frac{50\sqrt{3}}{50} = \sqrt{3}$$

$$\Rightarrow \theta = 60^\circ.$$

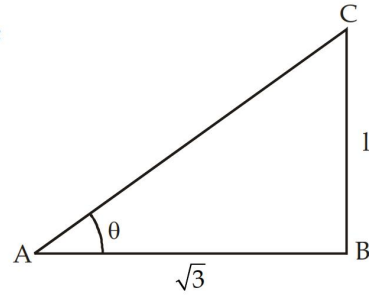
22. (b);



$$\frac{h}{30} = \tan 60^\circ = \sqrt{3}$$

$$\Rightarrow h = 30\sqrt{3} \text{ m.}$$

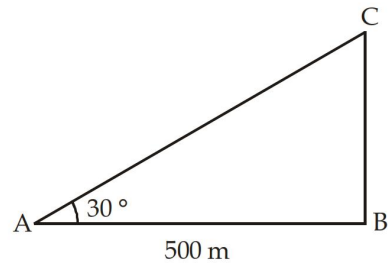
23. (b);



$$\tan \theta = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \theta = 30^\circ$$

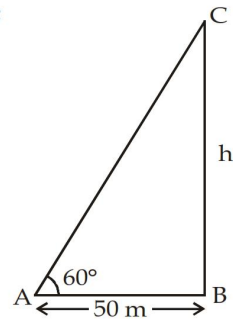
24. (c);



$$\frac{BC}{AB} = \tan 30^\circ$$

$$BC = 500 \tan 30^\circ = \frac{500}{\sqrt{3}} \text{ m.}$$

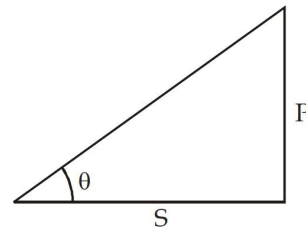
25. (d);



Height of the building = BC

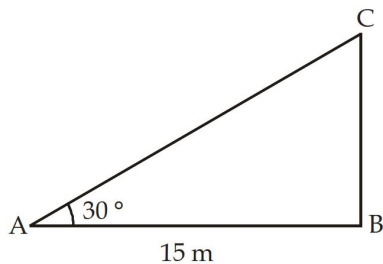
$$BC = 50 \tan 60^\circ = 50\sqrt{3} \text{ m.}$$

26. (c);



$$P = S \tan \theta = \frac{S}{\cot \theta}$$

27. (c);

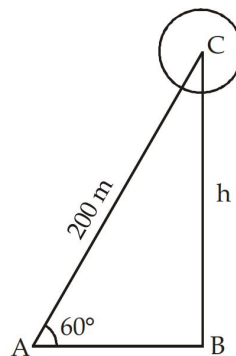


Length of the ladder = AC

$$\frac{AB}{AC} = \cos 30^\circ = \frac{\sqrt{3}}{2}$$

$$AC = \frac{2AB}{\sqrt{3}} \text{ m} = \frac{2 \times 15}{\sqrt{3}} \text{ m} = 10\sqrt{3} \text{ m}$$

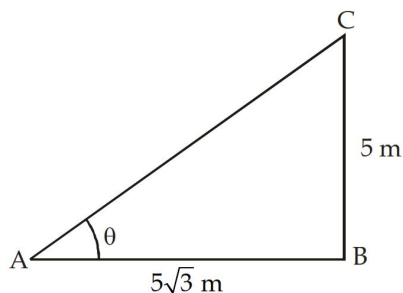
29. (a);



Height of the balloon from the ground = BC.

$$BC = 200 \sin 60^\circ = 200 \times \frac{\sqrt{3}}{2} = 100\sqrt{3} \text{ m} = 173.2 \text{ m}.$$

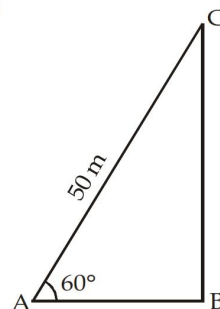
28. (d);



$$\tan \theta = \frac{BC}{AB} = \frac{5}{5\sqrt{3}} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \theta = 30^\circ$$

30. (b);



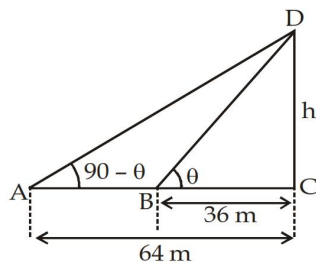
Height of the kite = BC = 50 sin 60°

$$= \frac{50\sqrt{3}}{2} = 25\sqrt{3} \text{ m}.$$

Moderate

Moderate

1. (b);



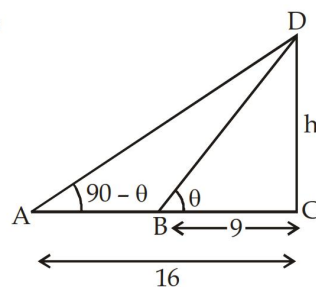
$$\tan \theta = \frac{h}{36}$$

$$\tan (90 - \theta) = \cot \theta = \frac{h}{64}$$

$$\tan \theta \times \cot \theta = 1$$

$$\Rightarrow \frac{h}{36} \times \frac{h}{64} = 1 \quad \Rightarrow h = 6 \times 8 = 48 \text{ cm}$$

2. (b);

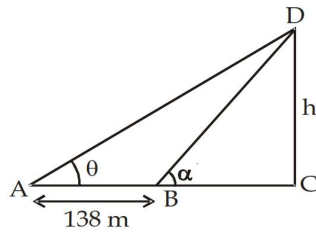


$$\frac{h}{9} = \tan \theta, \quad \frac{h}{16} = \tan (90 - \theta) = \cot \theta$$

$$\tan \theta \times \cot \theta = 1$$

$$\Rightarrow \frac{h}{9} \times \frac{h}{16} = 1 \quad \Rightarrow h = 3 \times 4 = 12 \text{ ft}.$$

3. (c);



$$\tan \theta = \frac{CD}{AC} = \frac{1}{5}$$

$$\Rightarrow AC = 5 CD$$

$$\tan \alpha = \frac{CD}{BC} = \sqrt{\sec^2 \alpha - 1}$$

$$\Rightarrow \frac{CD}{AC - 138} = \sqrt{\frac{193}{144} - 1} = \frac{7}{12}$$

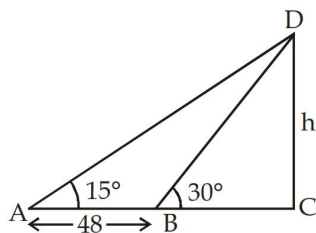
$$\Rightarrow AC = \frac{12}{7} CD + 138 \dots (ii)$$

From (i) and (ii)

$$\frac{12}{7} CD + 138 = 5 CD$$

$$\Rightarrow CD = 42 \text{ m}$$

4. (b);



Let the height of the tower be h.

$$BC = CD \cot 30^\circ$$

$$= \sqrt{3} h$$

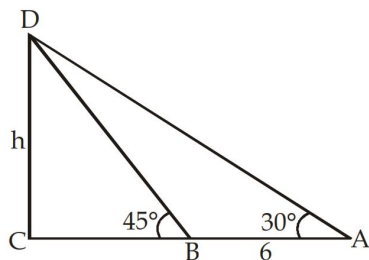
$$AC = CD \cot 15^\circ = h(2 + \sqrt{3})$$

$$AC - BC = 48 \text{ m}$$

$$\Rightarrow 2h = 48 \text{ m}$$

$$\Rightarrow h = 24 \text{ m.}$$

5. (a);



Let the height of the light-house is h.

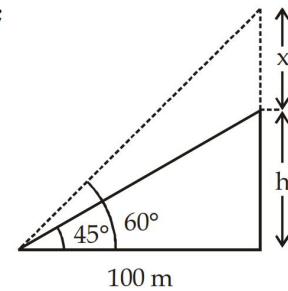
$$BC = h \cot 45^\circ = h$$

$$AC = h \cot 30^\circ = \sqrt{3} h$$

$$AB = AC - BC = 6 \text{ m}$$

$$\Rightarrow (\sqrt{3} - 1)h = 6 \Rightarrow h = 3(\sqrt{3} + 1) \text{ m}$$

6. (c);



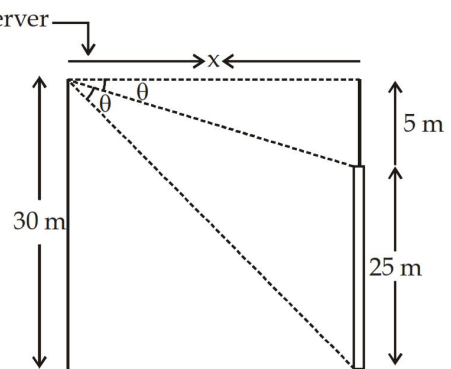
Let the height of the incomplete pillar be h and assume that it has to be increased by x m to complete it.

$$\tan 45^\circ = \frac{h}{100} = 1 \Rightarrow h = 100 \text{ m.}$$

$$\frac{h+x}{100} = \tan 60^\circ = \sqrt{3}$$

$$\Rightarrow h + x = 100\sqrt{3} \text{ m} \Rightarrow x = 100(\sqrt{3} - 1) \text{ m}$$

7. (b); observer



Let the distance of the observer from the top of the flag staff be x.

According to the question:

$$\tan \theta = \frac{5}{x}$$

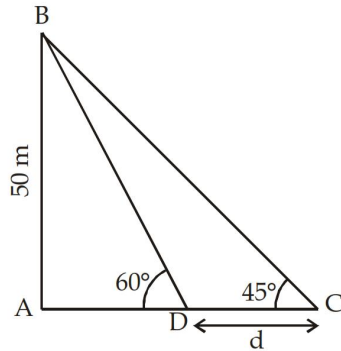
$$\tan 2\theta = \frac{30}{x}$$

$$\Rightarrow \tan 2\theta = 6 \tan \theta \Rightarrow \frac{2 \tan \theta}{(1 - \tan^2 \theta)} = 6 \tan \theta$$

$$\Rightarrow 1 - \tan^2 \theta = \frac{1}{3} \quad \Rightarrow \quad \tan \theta = \frac{\sqrt{2}}{\sqrt{3}} = \frac{5}{x}$$

$$\Rightarrow x = \frac{5\sqrt{3}}{\sqrt{2}}.$$

8. (b);



Let the distance between the objects be d.

$$\frac{AB}{AC} = \tan 45^\circ = 1$$

$$\Rightarrow AC = 50 \text{ m.}$$

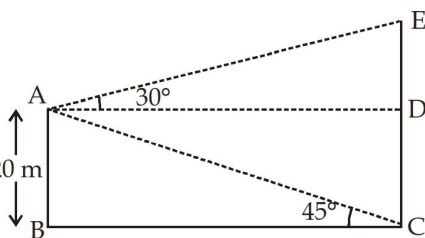
$$AD = AB \cot 60^\circ = \frac{AB}{\sqrt{3}}.$$

The distance between the objects

$$= d = AC - AD = 50 - \frac{50}{\sqrt{3}}$$

$$\Rightarrow d = 21 \text{ m (approx).}$$

9. (a);



$$\frac{AB}{BC} = \tan 45^\circ = 1$$

$$\Rightarrow BC = 20 \text{ m} \Rightarrow AD = 20 \text{ m}$$

$$\Rightarrow \tan 30^\circ = \frac{1}{\sqrt{3}} = \frac{DE}{20}$$

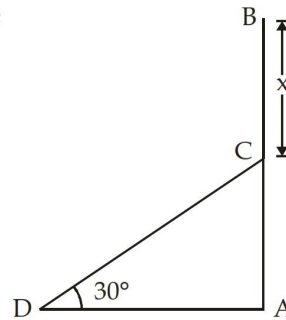
$$\Rightarrow DE = \frac{20}{\sqrt{3}} \text{ m}$$

$$CD = AB = 20 \text{ m}$$

$$\Rightarrow \text{The height of the second pillar} = CD + DE$$

$$= 20 + \frac{20}{\sqrt{3}} = \frac{20(\sqrt{3} + 1)}{\sqrt{3}}$$

10. (b);



Let CB be the part of the tree which breaks and takes the position CD.

$$\text{Let } CB = CD = x$$

$$AD = 10 \text{ m}$$

$$\angle ADC = 30^\circ$$

$$\frac{10}{x} = \cos 30^\circ = \frac{\sqrt{3}}{2} \Rightarrow x = \frac{20}{\sqrt{3}} \text{ m}$$

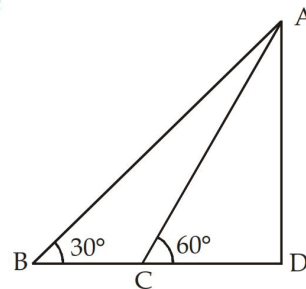
$$\frac{AC}{AD} = \tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$\Rightarrow AC = \frac{10}{\sqrt{3}} \text{ m.}$$

$$\text{Height of the tree} = AC + CB$$

$$= \frac{10}{\sqrt{3}} + \frac{20}{\sqrt{3}} = 10\sqrt{3} \text{ m.}$$

11. (b);



$$BD = a, CD = b$$

$$\frac{AD}{BD} = \tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \frac{AD}{a} = \frac{1}{\sqrt{3}}$$

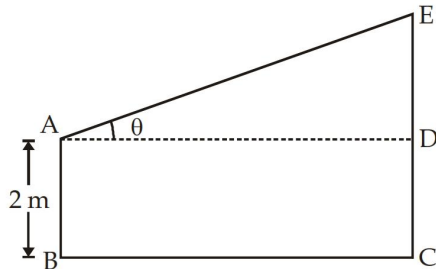
$$\frac{AD}{CD} = \tan 60^\circ = \sqrt{3}$$

$$\frac{AD}{b} = \sqrt{3}$$

$$\Rightarrow \frac{AD}{a} \times \frac{AD}{b} = \frac{1}{\sqrt{3}} \times \sqrt{3} = 1$$

$$\Rightarrow AD = \sqrt{ab}$$

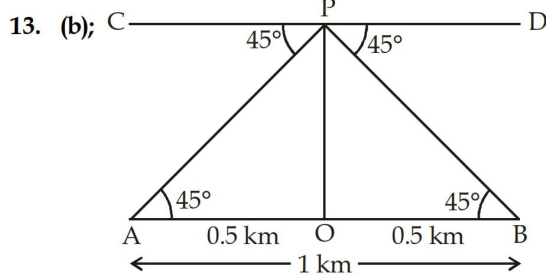
12. (b);



$$AD = BC = \frac{4}{\sqrt{3}} \text{ m}$$

$$DE = \frac{10}{3} - 2 = \frac{4}{3} \text{ m}$$

$$\tan \theta = \frac{DE}{AD} = \frac{\frac{4}{3}}{\frac{4}{\sqrt{3}}} = \frac{1}{\sqrt{3}} \Rightarrow \theta = 30^\circ$$



Let OP be the width of the river as shown in the diagram.

$$\therefore \angle OAP = \angle APC = 45^\circ$$

$$\angle OBP = \angle BPD = 45^\circ$$

$$\therefore \angle OAP = \angle OBP,$$

$$\therefore AP = BP$$

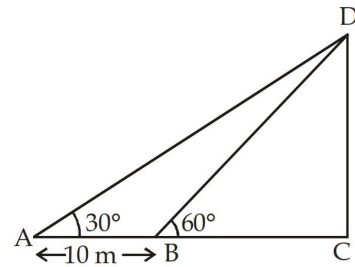
$\Rightarrow \triangle APB$ is an isosceles triangle OP is the distance between the two banks.

$$\Rightarrow OP \perp AB$$

$$\text{New, } \frac{OP}{OB} = \tan 45^\circ = 1$$

$$\Rightarrow OP = 1 \times 0.5 = 0.5 \text{ km.}$$

14. (a);



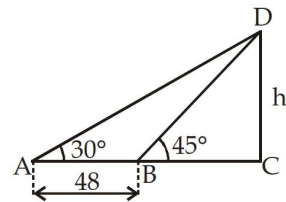
$$BC = h \cot 60^\circ = \frac{h}{\sqrt{3}}$$

$$AC = h \cot 30^\circ = \sqrt{3} h$$

$$AB = AC - BC = \sqrt{3} h - \frac{h}{\sqrt{3}} = 10 \text{ m}$$

$$\Rightarrow h = 5\sqrt{3} \text{ m.}$$

15. (a);



Let the height of the tower be h.

$$BC = h \cot 45^\circ = h$$

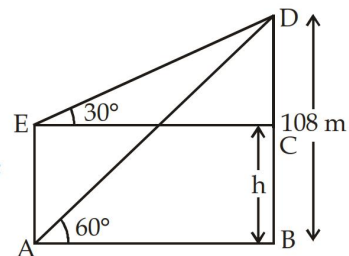
$$AC = h \cot 30^\circ = \sqrt{3} h$$

$$AB = (\sqrt{3} - 1) h = 48 \text{ m}$$

$$h = \frac{48(\sqrt{3} + 1)}{2}$$

$$h = 24(1 + \sqrt{3}) \text{ m.}$$

16. (b);



Let the height of the other post be h m.

$$\Rightarrow BC = h \text{ m.}$$

$$\Rightarrow CD = (108 - h) \text{ m.}$$

$$AB = BD \cot 60^\circ = \frac{BD}{\sqrt{3}} = \frac{108}{\sqrt{3}}$$

$$EC = AB = \frac{108}{\sqrt{3}} \text{ m}$$

$$\frac{CD}{EC} = \tan 30^\circ = \frac{1}{\sqrt{3}}$$

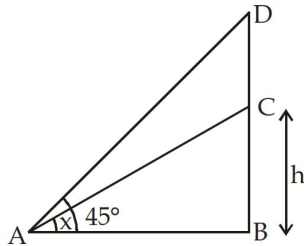
$$\frac{108 - h}{\frac{108}{\sqrt{3}}} = \frac{1}{\sqrt{3}}$$

$$\frac{\sqrt{3}(108 - h)}{108} = \frac{1}{\sqrt{3}}$$

$$108 - h = \frac{108}{3}$$

$$h = 108 - \frac{108}{3} = \frac{2 \times 108}{3} = 72 \text{ m.}$$

17. (b);



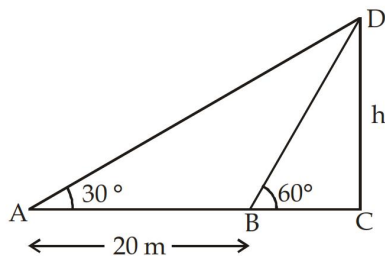
$$AB = h \cot x$$

$$\frac{BD}{AB} = \tan 45^\circ = 1$$

$$\Rightarrow BD = h \cot x$$

$$\text{The height of the chimney} = CD = BD - h \\ = h \cot x - h.$$

18. (c);



Let the height of the tower be h .

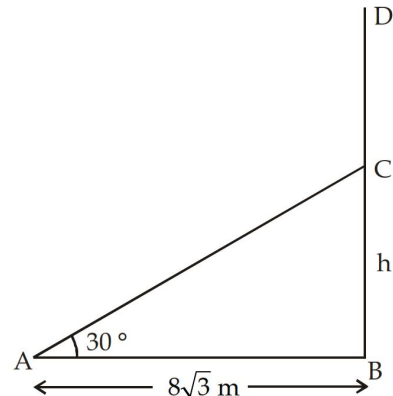
$$AC = h \cot 30^\circ = \sqrt{3} h$$

$$BC = h \cot 60^\circ = \frac{h}{\sqrt{3}}$$

$$AB = AC - BC = \left(\sqrt{3} - \frac{1}{\sqrt{3}} \right) h = 20 \text{ m.}$$

$$\Rightarrow h = 10\sqrt{3} \text{ m.}$$

19. (c);



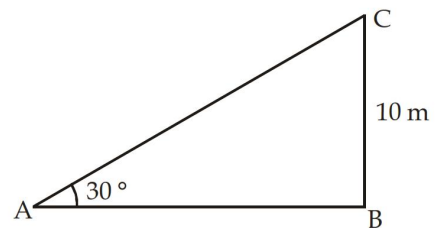
$$BC = AB \tan 30^\circ = 8\sqrt{3} \times \frac{1}{\sqrt{3}}$$

$$\frac{BC}{AC} = \sin 30^\circ = \frac{1}{2}$$

$$AC = 2 BC = 16 = CD$$

$$\text{Height of the post} = BD = BC + CD \\ = 8 + 16 = 24 \text{ m}$$

20. (c);



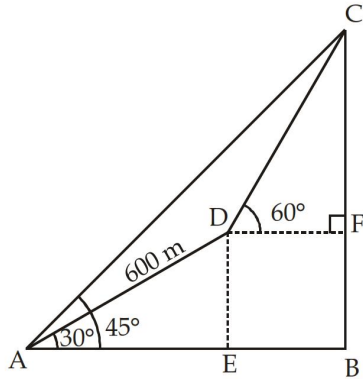
$$\frac{AB}{BC} = \cot 30^\circ = \sqrt{3}$$

$$AB = 10\sqrt{3} \text{ m}$$

So, the distance of the foot of the ladder from the wall is $10\sqrt{3} \text{ m}$

Difficult

1. (c);



$$\begin{aligned}\angle DCF &= 180^\circ - (90^\circ + 60^\circ) \\ &= 180^\circ - 150^\circ = 30^\circ \\ \angle ACB &= 180^\circ - (90^\circ + 45^\circ) = 45^\circ \\ \angle ACD &= \angle ACB - \angle DCF = 45^\circ - 30^\circ = 15^\circ \\ \angle DAC &= 45^\circ - 30^\circ = 15^\circ\end{aligned}$$

In $\triangle ACD$: $\angle ACD = \angle DAC = 15^\circ$

$\Rightarrow DC = AD = 600$ m.

In $\triangle ADE$:

$$DE = 600 \sin 30^\circ = 600 \times \frac{1}{2} = 300 \text{ m.}$$

$BF = DE = 300$ m.

in $\triangle DFC$:

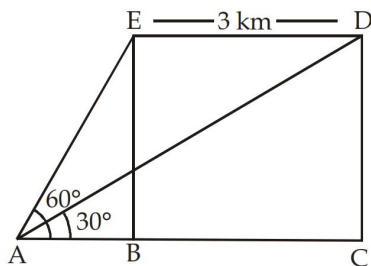
$CF = CD \sin 60^\circ$

$$CF = 600 \times \frac{\sqrt{3}}{2} = 300\sqrt{3} \text{ m.}$$

Height of the mountain =

$$BC = BF + FC = 300 + 300\sqrt{3} = 300(1 + \sqrt{3}) \text{ m.}$$

2. (c);



Distance covered in 15 seconds at

$$720 \text{ kmph} = 720 \times \frac{5}{18} \times 15 = 3000 \text{ m} = 3 \text{ km.}$$

$$AC = h \cot 30^\circ = \sqrt{3} h$$

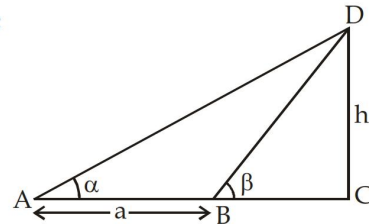
$$AB = h \cot 60^\circ = \frac{h}{\sqrt{3}}.$$

$$BC = AC - AB = 3 \text{ km.}$$

$$\sqrt{3} h - \frac{h}{\sqrt{3}} = 3$$

$$\Rightarrow h = \frac{3}{2} \sqrt{3} \text{ km} = \frac{3000}{2} \times 1.732 \text{ m} = 2598 \text{ m.}$$

3. (d);



Let the height of the tower be h . $AC = h \cot \alpha$

$$AC = h \cot \alpha$$

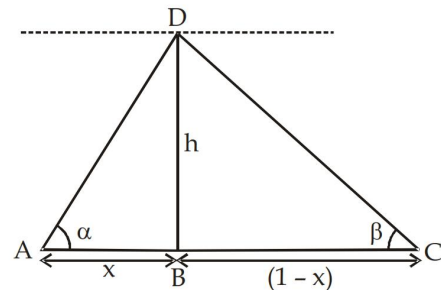
$$BC = h \cot \beta$$

$$AB = AC - BC = a$$

$$\Rightarrow h(\cot \alpha - \cot \beta) = a$$

$$\Rightarrow h = \frac{a}{\cot \alpha - \cot \beta} = \frac{a \tan \alpha \tan \beta}{\tan \beta - \tan \alpha}$$

4. (c);



Let A and C be positions of two consecutive milestones

$$\Rightarrow AC = 1 \text{ km.}$$

$$\text{Let } AB = x \Rightarrow BC = (1 - x)$$

Let the height of the aeroplane be h .

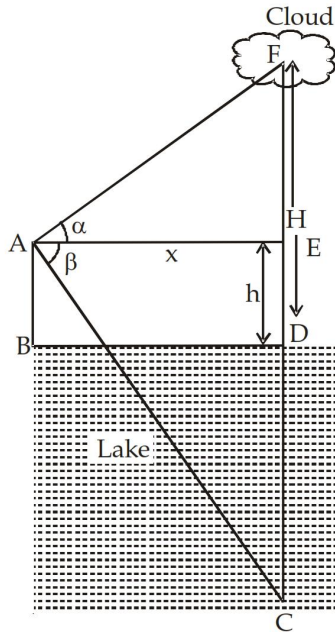
$$x = h \cot \alpha$$

$$1 - x = h \cot \beta$$

$$h(\cot \alpha + \cot \beta) = 1$$

$$h = \frac{1}{\cot \alpha + \cot \beta} = \frac{\tan \alpha \tan \beta}{\tan \beta + \tan \alpha}.$$

5. (d);



Let the height of the cloud above the lake be H m.

Let AE = x m.

$x = (H - h) \cot \alpha$ (From $\triangle AEF$).

in $\triangle ACE$:

$x = (H + h) \cot \beta$

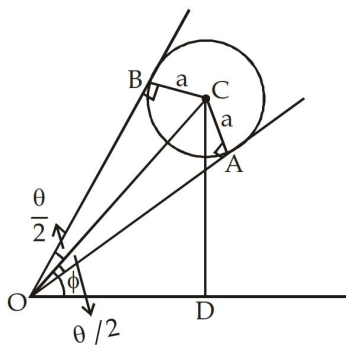
$$\Rightarrow \frac{H - h}{\tan \alpha} = \frac{H + h}{\tan \beta}$$

$$H \tan \alpha + h \tan \alpha = H \tan \beta - h \tan \beta$$

$$h(\tan \alpha + \tan \beta) = H(\tan \beta - \tan \alpha)$$

$$\Rightarrow H = \frac{h(\tan \alpha + \tan \beta)}{\tan \beta - \tan \alpha}.$$

6. (c);



OA and OB are tangents to the spherical balloon.

$\Rightarrow CA \perp OA$ i.e., $\angle OAC = 90^\circ$.

Similarly, $CB \perp OB$ i.e., $\angle OBC = 90^\circ$

in $\triangle OAC$,

$$\operatorname{cosec} \frac{\theta}{2} = \frac{OC}{a}$$

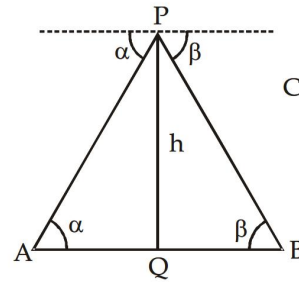
$$\Rightarrow OC = a \operatorname{cosec} \frac{\theta}{2}$$

in $\triangle ODC$,

$$CD = OC \sin \phi, \quad CD = a \operatorname{cosec} \frac{\theta}{2} \sin \phi$$

So, height of the centre of ball is: $a \sin \phi \operatorname{cosec} \frac{\theta}{2}$.

7. (a);



The height of the aeroplane as shown in the figure = PQ = h km.

$$\frac{h}{AQ} = \tan \alpha$$

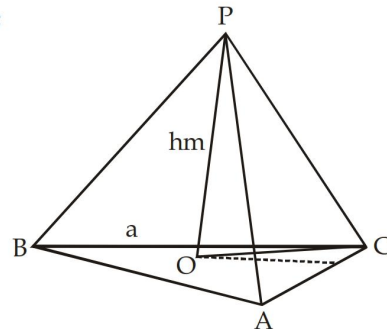
$$\Rightarrow AQ = \frac{h}{\tan \alpha} \Rightarrow \frac{h}{BQ} = \tan \beta$$

$$\Rightarrow BQ = \frac{h}{\tan \beta}$$

$$AQ + BQ = 1$$

$$\Rightarrow \frac{h}{\tan \alpha} + \frac{h}{\tan \beta} = 1 \Rightarrow h = \frac{\tan \alpha \tan \beta}{\tan \alpha + \tan \beta}$$

8. (b);



Let ABC be the equilateral triangle Let O be the centre of $\triangle ABC$ and let OP be the h m high tower. Now, each side of the equilateral triangle ABC subtends an angle of 60° at the top of the tower.

$\Rightarrow$ Each of $\triangle ABP$, $\triangle ACP$ and $\triangle BCP$ is equilateral.

$$\Rightarrow AP = BP = CP = a$$

$$OB = \frac{a}{2} \sec 30^\circ$$

In $\triangle OBP$:

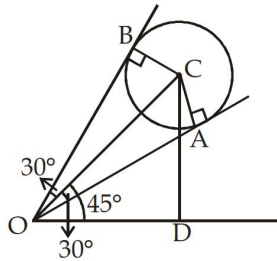
$$\Rightarrow h^2 + \left(\frac{a}{2} \sec 30^\circ\right)^2 = a^2$$

$$\Rightarrow h^2 + \frac{a^2}{3} = a^2$$

$$\Rightarrow h^2 = \frac{2a^2}{3}$$

$$\Rightarrow 2a^2 = 3h^2.$$

9. (c);



OA and OB are tangents to the spherical balloon. similarly, $CB \perp OB$, i.e., $\angle OBC = 90^\circ$.
in $\triangle OAC$:

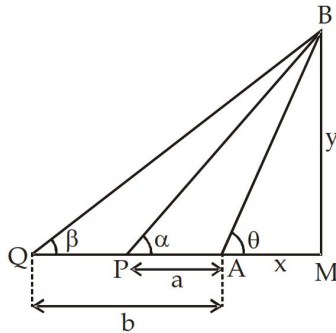
$$\frac{OC}{10} = \operatorname{Cosec} 30^\circ$$

$$OC = 20 \text{ m.}$$

in $\triangle ODC$:

$$CD = OC \sin 45^\circ = \frac{20}{\sqrt{2}} \text{ m} = 10\sqrt{2} \text{ m.}$$

10. (c);



Let AB be the tower inclined at an angle θ to the horizontal.

Let $AM = x$ and $BM = y$.

$$\cot \alpha = \frac{a+x}{y}$$

$$\Rightarrow a+x = y \cot \alpha$$

$$\cot \beta = \frac{b+x}{y}$$

$$\Rightarrow b+x = y \cot \beta$$

Subtract (ii) from (i):

$$a-b = y(\cot \alpha - \cot \beta).$$

$$\Rightarrow y = \frac{a-b}{\cot \alpha - \cot \beta}.$$

From (i):

$$x = y \cot \alpha - a$$

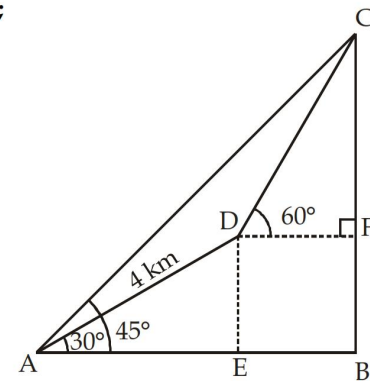
$$= \frac{(a-b) \cot \alpha}{\cot \alpha - \cot \beta} - a = \frac{a \cot \beta - b \cot \alpha}{\cot \alpha - \cot \beta}$$

in $\triangle ABM$:

$$\cot \theta = \frac{x}{y} = \frac{\frac{a \cot \beta - b \cot \alpha}{\cot \alpha - \cot \beta}}{\frac{a-b}{\cot \alpha - \cot \beta}}$$

$$\Rightarrow \cot \theta = \frac{b \cot \alpha - a \cot \beta}{b-a}$$

11. (a);



In $\triangle ABC$: $\angle ACB = 180^\circ - (90^\circ + 45^\circ) = 45^\circ$

In $\triangle DCF$: $\angle DCF = 180^\circ - (90^\circ + 60^\circ) = 30^\circ$

$$\Rightarrow \angle ACD = \angle ACB - \angle DCF = 45^\circ - 30^\circ = 15^\circ$$

$$\angle CAD = \angle CAB - \angle DAE = 45^\circ - 30^\circ = 15^\circ.$$

$$\Rightarrow \angle ACD = \angle CAD$$

$$\Rightarrow DC = AD = 4 \text{ km} \quad [\text{isosceles triangle}]$$

$$CF = CD \sin 60^\circ = \frac{\sqrt{3}}{2} \cdot 4 = 2\sqrt{3} \text{ km}$$

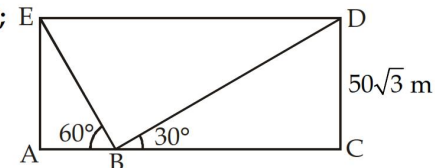
in $\triangle AED$:

$$DE = AD \sin 30^\circ = \frac{4}{2} = 2 \text{ km.}$$

Height of the mountain = $CF + FB$

$$= 2\sqrt{3} + 2 = 2(1 + \sqrt{3}) \text{ km.}$$

12. (d);



... (i)

$$BC = 50\sqrt{3} \cot 30^\circ = 50\sqrt{3} \times \sqrt{3} = 150 \text{ m}$$

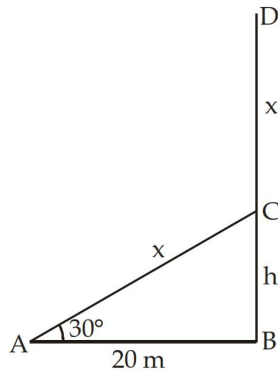
$$AB = 50\sqrt{3} \cot 60^\circ = 50 \text{ m.}$$

$$AC = AB + BC = 50 + 150 = 200 \text{ m.}$$

So, the bird covers 200 m in 2 minutes speed =

$$= \frac{200}{2 \times 60} \text{ m/sec} = \frac{200}{120} \times \frac{18}{5} \text{ km/h} = 6 \text{ km/h.}$$

13. (a);



According to the figure :

$$h = 20 \tan 30^\circ = \frac{20}{\sqrt{3}} \text{ m}$$

$$\frac{h}{x} = \sin 30^\circ = \frac{1}{2}$$

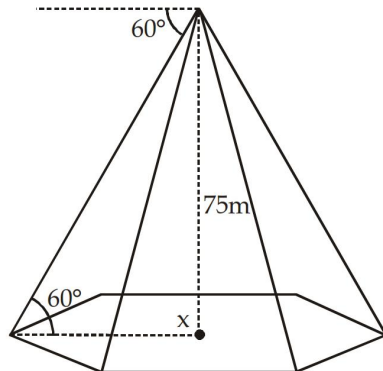
$$\frac{20}{\sqrt{3}x} = \frac{1}{2}$$

$$\Rightarrow x = \frac{40}{\sqrt{3}} \text{ m.}$$

Height of the pole before it was broken = $h + x$

$$= \frac{20}{\sqrt{3}} + \frac{40}{\sqrt{3}} = \frac{60}{\sqrt{3}} = 20\sqrt{3} \text{ m.}$$

14. (c);



Angle of elevation = Angle of depression

So, each slant edge makes an angle of 60° with the horizontal. Let the distance between a vertex and centre of the hexagon be x .

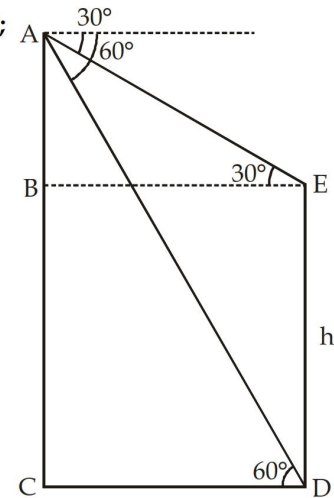
$$\tan 60^\circ = \frac{75}{x} = \sqrt{3}$$

$$x = 25\sqrt{3} \text{ m.}$$

$\therefore$ distance between the centre and the vertices of a hexagon is equal to the length of the edges,

$\therefore$ Length of each side of the hexagon = $25\sqrt{3} \text{ m.}$

15. (b);



$$\frac{108}{CD} = \tan 60^\circ = \sqrt{3}$$

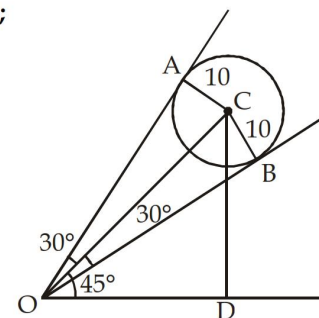
$$CD = \frac{108}{\sqrt{3}} = \frac{108\sqrt{3}}{3} = 36\sqrt{3} \text{ m.}$$

$$BE = CD = 36\sqrt{3} \text{ m.}$$

$$AB = BE \tan 30^\circ = 36\sqrt{3} \times \frac{1}{\sqrt{3}} = 36 \text{ m.}$$

$$\Rightarrow \text{Height of the second pillar} = DE = BC = AC - AB = 108 - 36 = 72 \text{ m.}$$

16. (d);



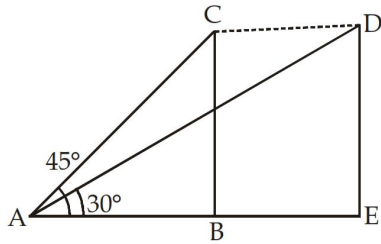
Let the height of the centre of the balloon be h .

According to the figure:

$$OC = 10 \operatorname{cosec} 30^\circ = 10 \times 2 = 20$$

$$h = OC \sin 45^\circ = \frac{20}{\sqrt{2}} = 10\sqrt{2} \text{ ft.}$$

17. (a);



$$CB = DE = 2500 \text{ m}$$

$$\frac{CB}{AB} = \tan 45^\circ = 1$$

$$\Rightarrow AB = CB = 2500 \text{ m}$$

In $\triangle ADE$:

$$\frac{AE}{DE} = \cot 30^\circ = \sqrt{3}$$

$$\Rightarrow AE = 2500\sqrt{3} \text{ m.}$$

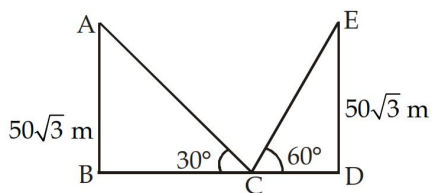
$$BE = AE - AB = 2500\sqrt{3} - 2500 = 2500(\sqrt{3} - 1) \text{ m}$$

So the aeroplane covers $2500(\sqrt{3} - 1)$ meters in 15 seconds.

$$\Rightarrow \text{speed} = \frac{2500}{15}(\sqrt{3} - 1) \text{ m/s.}$$

$$= \frac{2500}{15}(\sqrt{3} - 1) \times \frac{18}{5} \text{ km/h} = 440 \text{ kmph (approx).}$$

18. (d);



$$BC = 50\sqrt{3} \cot 30^\circ$$

$$= 50\sqrt{3} \times \sqrt{3} = 150 \text{ m}$$

$$CD = 50\sqrt{3} \cot 60^\circ$$

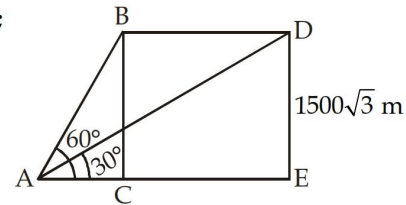
$$= 50\sqrt{3} \times \frac{1}{\sqrt{3}} = 50 \text{ m}$$

$$BD = BC + CD = 200 \text{ m.}$$

So, the bird covers a distance of 200 m in 2 minute i.e., 120 seconds

$$\begin{aligned} \text{Speed} &= \frac{200}{120} \text{ m/s} = \frac{200}{120} \times \frac{18}{5} \text{ km/h} \\ &= 6 \text{ km/h.} \end{aligned}$$

19. (b);



$$AC = 1500\sqrt{3} \times \frac{1}{\sqrt{3}} = 1500 \text{ m.}$$

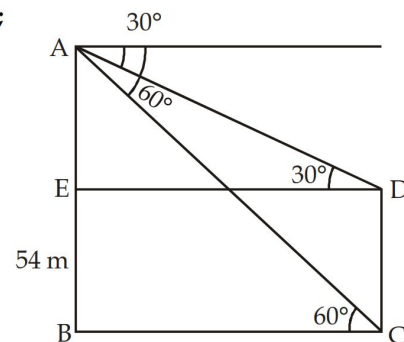
$$AE = 1500\sqrt{3} \cot 30^\circ = 1500\sqrt{3} \times \sqrt{3} = 4500 \text{ m.}$$

$$CE = AE - AC = 4500 - 1500 = 3000 \text{ m.}$$

So, the aeroplane cover 3000 m in 15 seconds.

$$\text{speed} = \frac{3000}{15} = 200 \text{ m/s}$$

20. (b);



In $\triangle ABC$

$$BC = AB \cot 60^\circ$$

$$= 54 \times \frac{1}{\sqrt{3}} = \frac{54\sqrt{3}}{3} = 18\sqrt{3} \text{ m}$$

$$DE = BC = 18\sqrt{3} \text{ m}$$

$$AE = DE \tan 30^\circ = 18\sqrt{3} \times \frac{1}{\sqrt{3}} = 18 \text{ m}$$

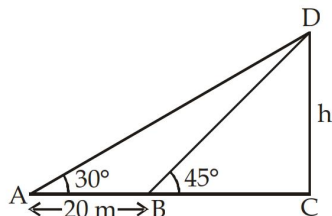
The height of the other

$$\text{temple} = CD = BE = AB - AE$$

$$= (54 - 18) \text{ m} = 36 \text{ m}$$

Previous Year (Memory Based)

1. (d);



Let the height of the pillar be h .

$$AC = h \cot 30^\circ = \sqrt{3} h$$

$$BC = h \cot 45^\circ = h$$

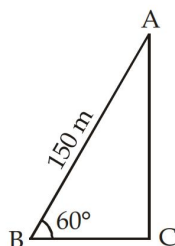
$$AB = AC - BC = 20 \text{ m}$$

$$\Rightarrow (\sqrt{3} - 1)h = 20 \text{ m}$$

$$h = \frac{20}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1}$$

$$= 10(\sqrt{3} + 1) \text{ m.}$$

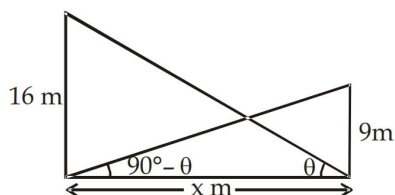
2. (b);



Height of the kite from the ground = $AC = 150 \sin 60^\circ$

$$= 75\sqrt{3} \text{ m.}$$

3. (c);



$$x = 16 \cot \theta$$

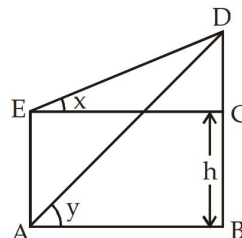
$$x = 9 \cot (90^\circ - \theta) = 9 \tan \theta$$

$$x = 9 \times \frac{16}{x}$$

$$x^2 = 144$$

$$x = 12$$

4. (c);



$$BD = AB \tan y.$$

$$DC = EC \tan x = AB \tan x$$

$$BC = h = BD - DC$$

$$h = AB \tan y - AB \tan x$$

$$\Rightarrow AB = \frac{h}{\tan y - \tan x}$$

$$\Rightarrow DC = \frac{h \tan x}{\tan y - \tan x}$$

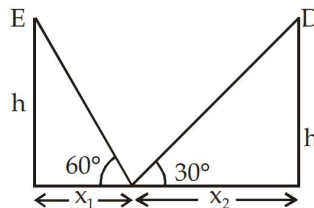
So, the height of the building

$$= BD = BC + CD = h + CD$$

$$= h + \frac{h \tan x}{\tan y - \tan x} = \frac{h \tan y}{\tan y - \tan x}$$

$$= \frac{h \cot x}{\cot x - \cot y}$$

5. (a);



$$x_1 = h \cot 60^\circ = \frac{h}{\sqrt{3}}$$

$$x_2 = h \cot 30^\circ = \sqrt{3} h$$

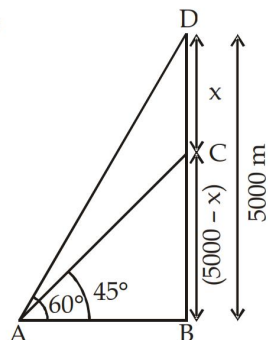
$$x_1 + x_2 = 100 \text{ m}$$

$$\Rightarrow \frac{h}{\sqrt{3}} + \sqrt{3} h = 100$$

$$\Rightarrow 4h = 100\sqrt{3}.$$

$$\Rightarrow h = 25\sqrt{3} \text{ m.}$$

6. (C);



$$\tan 45^\circ = 1 = \frac{(5000 - x)}{AB}$$

$$\Rightarrow AB = (5000 - x).$$

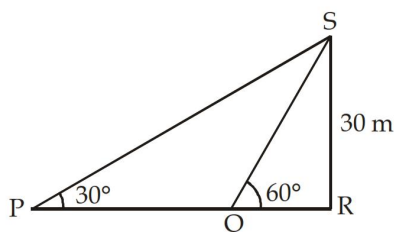
In $\triangle ABD$

$$\tan 60^\circ = \sqrt{3} = \frac{5000}{5000 - x}$$

$$\Rightarrow 5000\sqrt{3} - \sqrt{3}x = 5000$$

$$\Rightarrow x = 5000 \left(1 - \frac{1}{\sqrt{3}} \right) \text{ m.}$$

7. (d);



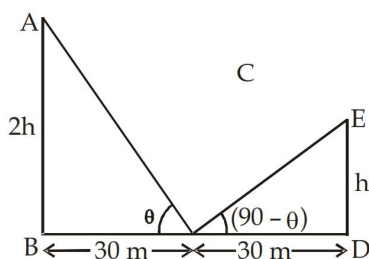
$$QR = 30 \cot 60^\circ = \frac{30}{\sqrt{3}} \text{ m}$$

$$PR = 30 \cot 30^\circ = 30\sqrt{3} \text{ m.}$$

The distance he moves = $PR - QR$

$$= \left(30\sqrt{3} - \frac{30}{\sqrt{3}} \right) \text{ m} = 30 \left(\frac{2}{\sqrt{3}} \right) = 20\sqrt{3} \text{ m.}$$

8. (d);



Let the height of one of the poles be h .

So, height of the other pole = $2h$

$$\tan \theta = \frac{2h}{30}$$

$$\tan (90 - \theta) = \cot \theta = \frac{h}{30}$$

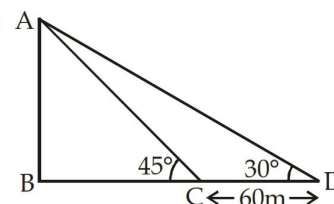
$$\Rightarrow \frac{2h}{30} \times \frac{h}{30} = 1$$

$$\Rightarrow h^2 = 15 \times 30$$

$$\Rightarrow h = 15\sqrt{2} \text{ m}$$

$$\text{So, } 2h = 30\sqrt{2} \text{ m.}$$

9. (c);



$$BD = AB \cot 30^\circ$$

$$BD = \sqrt{3} AB$$

$$BC = AB \cot 45^\circ = AB$$

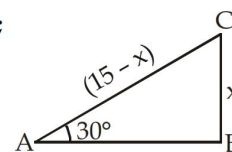
$$CD = BD - BC$$

$$= \sqrt{3} AB - AB = AB (\sqrt{3} - 1) = 60$$

$$\Rightarrow \text{Height of the tower} = AB = \frac{60}{\sqrt{3} - 1}$$

$$= 30(1 + \sqrt{3}) \text{ m.}$$

10. (b);



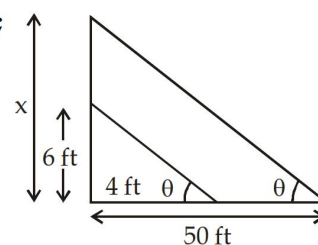
$$\sin 30^\circ = \frac{1}{2} = \frac{x}{15 - x}$$

$$\Rightarrow 2x = 15 - x$$

$$\Rightarrow x = 5$$

So, the post is broken at a height of 5 ft.

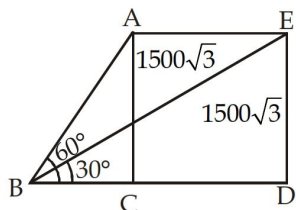
11. (b);



$$\tan \theta = \frac{6}{4} = \frac{3}{2} = \frac{x}{50}$$

$$\Rightarrow x = 75 \text{ ft}$$

12. (b);



$$BC = 1500\sqrt{3} \cot 60^\circ$$

$$= 1500\sqrt{3} \times \frac{1}{\sqrt{3}} = 1500$$

$$BD = 1500\sqrt{3} \cot 30^\circ$$

$$= 4500 \text{ m.}$$

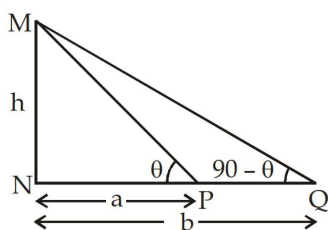
$$CD = BD - BC$$

$$= 4500 - 1500 = 3000 \text{ m.}$$

So, the aeroplane covers 3000 m in 15 seconds.

$$\text{Speed} = \frac{3000}{15} = 200 \text{ m/sec.}$$

13. (a);



Let the height of the tower be h.

$$\tan \theta = \frac{h}{a}$$

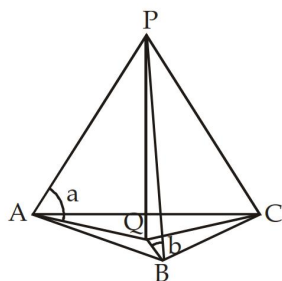
$$\tan (90 - \theta) = \cot \theta = \frac{h}{b}$$

$$\Rightarrow \frac{h}{a} \times \frac{h}{b} = \tan \theta \times \cot \theta = 1$$

$$\Rightarrow h^2 = ab$$

$$\Rightarrow h = \sqrt{ab}.$$

14. (b);



According to the questions:

$$\angle a = \angle b = \angle c$$

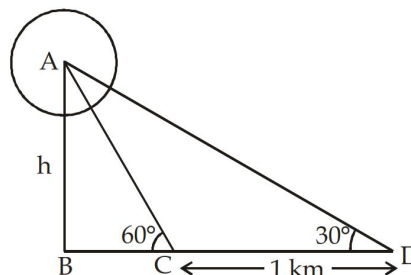
$$\Rightarrow \tan a = \tan b = \tan c$$

$$\Rightarrow \frac{PQ}{AQ} = \frac{PQ}{BQ} = \frac{PQ}{CQ}.$$

$$\Rightarrow AQ = BQ = CQ$$

$\Rightarrow$ the foot of the pole (Q) is at the circumcentre of $\triangle ABC$

15. (a);



Let the height of the balloon above the ground be h.

$$BC = h \cot 60^\circ = \frac{h}{\sqrt{3}}$$

$$BD = h \cot 30^\circ = \sqrt{3} h$$

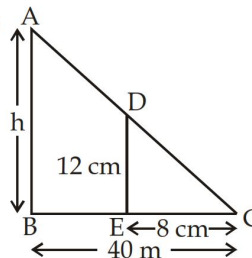
$$CD = BD - BC = 1 \text{ km.}$$

$$\Rightarrow \sqrt{3} h - \frac{h}{\sqrt{3}} = 1$$

$$\Rightarrow h \left(\sqrt{3} - \frac{1}{\sqrt{3}} \right) = 1$$

$$\Rightarrow h = \frac{\sqrt{3}}{2} \text{ km.}$$

16. (b);

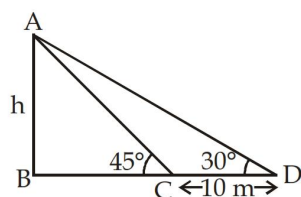


Let the height of the tower be h. According to the question:

$$\frac{12}{8} = \frac{h}{4000}$$

$$\Rightarrow h = \frac{12 \times 4000}{8} \text{ cm} = 60 \text{ m}$$

17. (c);



Let the height of the tower be h .

$$BC = h \cot 45^\circ$$

$$\Rightarrow BC = h$$

$$BD = h \cot 30^\circ = \sqrt{3} h$$

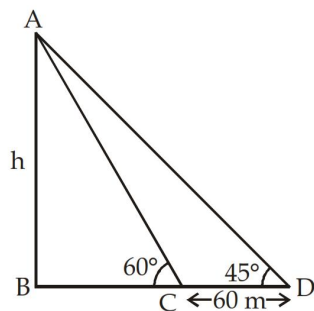
$$CD = 10\text{ m} = BD - BC$$

$$\Rightarrow \sqrt{3} h - h = 10$$

$$\Rightarrow h = \frac{10}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1}$$

$$\Rightarrow 5(1 + \sqrt{3})\text{ m}$$

18. (c);



Let the height of the tower be h m.

$$BC = h \cot 60^\circ = \frac{h}{\sqrt{3}}$$

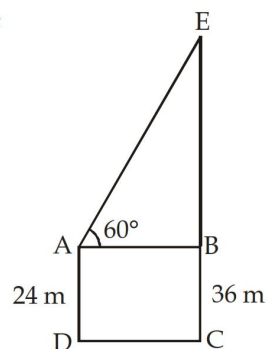
$$BD = h \cot 45^\circ = h$$

$$CD = BD - BC = 60\text{ m.}$$

$$\Rightarrow h - \frac{h}{\sqrt{3}} = 60$$

$$\Rightarrow h = 30(3 + \sqrt{3})$$

19. (b);



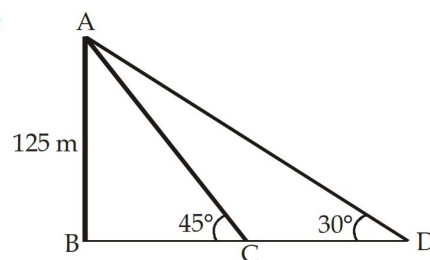
$$BE = 36 - 24 = 12\text{ m}$$

$$\frac{AE}{BE} = \operatorname{cosec} 60^\circ = \frac{2}{\sqrt{3}}$$

$$= AE = 12 \times \frac{2}{\sqrt{3}}\text{ m} = 8\sqrt{3}\text{ m.}$$

So, length of the wire is $8\sqrt{3}$ m.

20. (b);



$$BC = 125 \cot 45^\circ = 125\text{ m.}$$

$$BD = 125 \cot 30^\circ = 125\sqrt{3}\text{ m}$$

So, the distance between the objects

$$= CD = BD - BC = 125(\sqrt{3} - 1)\text{ m.}$$